Project Plan Group 6

1. Project description

The goal of this project is to design and implement a distributed Vehicle-to-Infrastructure (V2I) system using a microservices architecture deployed on Kubernetes (GKE).
The system simulates assisted driving scenarios with communication between vehicle ECUs and a backend.

Project goals:

- Develop a distributed microservice system for vehicle communication
- Deploy services using containerization and orchestration (GKE)
- Implement real-time data exchange via message broker/API gateway
- Build a Storefront web application to visualize vehicle behavior
- Simulate 3 driving scenarios (constant speed, emergency braking, dynamic speed change)
- Ensure scalability, fault tolerance and modularity

Project Non-goals:

- Real vehicle integration
- Performance optimization beyond a stable demo

2. Project Organization Chart

Role	Name	Backup	Responsibilities
Backend & Cloud Engineer	Lubomira Smolarova	Thomas Czech	Kubernetes, GCP setup, message broker, database
Microservices Developer	Martin Resch	Lubomira Smolarova	WHEREAMI, SONAR, BRAKENOW, UTRACKED, ORCHESTRATOR services
Frontend & Integration Engineer	Thomas Czech	Martin Resch	Storefront UI, API Gateway

3. Work Breakdown Structure (WBS)

Nr.	Task	Description	Responsible (Lead)	Milestone
1	**Projektmanagement & Planning**	Create a Project Plan, Define Milestones, GCP-Budget, Git Setup	Lubomira Smolarova	M1
2	**Architecture & Design**	Logical View & Physical View, API-Definitions, EER-Diagrams	Lubomira Smolarova	M2
3	**Infrastructure**	GCP Cluster Setup, Kubernetes	Lubomira Smolarova	M3
4	**Messaging Service**	Mom & Spider	Martin Resch	M3
5	**BackendServices (Data Persistence)**	WHEREAMI, UTRACKED	Martin Resch	M4
6	**Backend Services (Logic)**	SONAR, BRAKENOW, ORCHESTRATOR	Martin Resch	M4
7	**Front End**	Storefront	Thomas Czech	M4
8	**Simulation**	Simulator for the 3 scenarios	Thomas Czech	M5
9	**Testing**	Write Unit and Integration tests	Thomas Czech	M5
10	**Documentation**		All	M6

4. Effort Estimation

The following table lists the estimated time, per work package. The amount of work per person is oriented on the 3 ECTS (75-90h) this course is rated. We estimate that each team member spends 85h on the project.

WP Nr.	Time (h)
1	15h
2	25h
3	35h
4	20h

5	25h
6	35h
7	30h
8	35h
9	20h
10	3x 5h
Total	=255h

5. Resource & Milestone Planning

5.1 Milestones

ID	Milestone	Date (Deadline)	Description
M1	Project Kick-off	12.04.2026	Finalization of project plan
M2	Design Freeze	26.04.2026	Architecture & API-Spezification is finalised
M3	Infrastructure ready	10.05.2026	GKE Cluster is operational, Message middleware is setup
M4	Feature Complete	24.05.2026	Microservices & Storefront are functional
M5	Simulation & Testing	31.05.2026	Scenarios & Tests are finished
M6	Project finished	07.06.2026	All deliverables are finished and submitted via tuwel

5.2 GCP-Budget

Google Kubernetes Engine costs 0.10USD per hour, but this will be substituted for the first month.
Also an ‘e2-medium’ node costs 0.033 USD per hour. After also adding the external IP and estimating a low amount of disc space for the databases. It should cost around 80-100USD. This results in a buffer of 50USD.

Item	**Estimated Cost Range**
GKE Management Fee	40-80USD (0.10USD /h)
Storage and persistent disks	5 USD
Network/ IP-Lease	5-10 USD
e2-medium	5 USD (0.033 USD /h)

6. Engineering Planning

- Containerization: Docker
- Orchestration: Kubernetes (GKE)
- Cloud Provider: Google Cloud Platform
- Message Broker: RabbitMQ / Cloud Pub/Sub
- Backend: C# (.Net)
- Frontend: React Framework
- Database: Firestore / Cloud SQL
- Version Control: GitHub